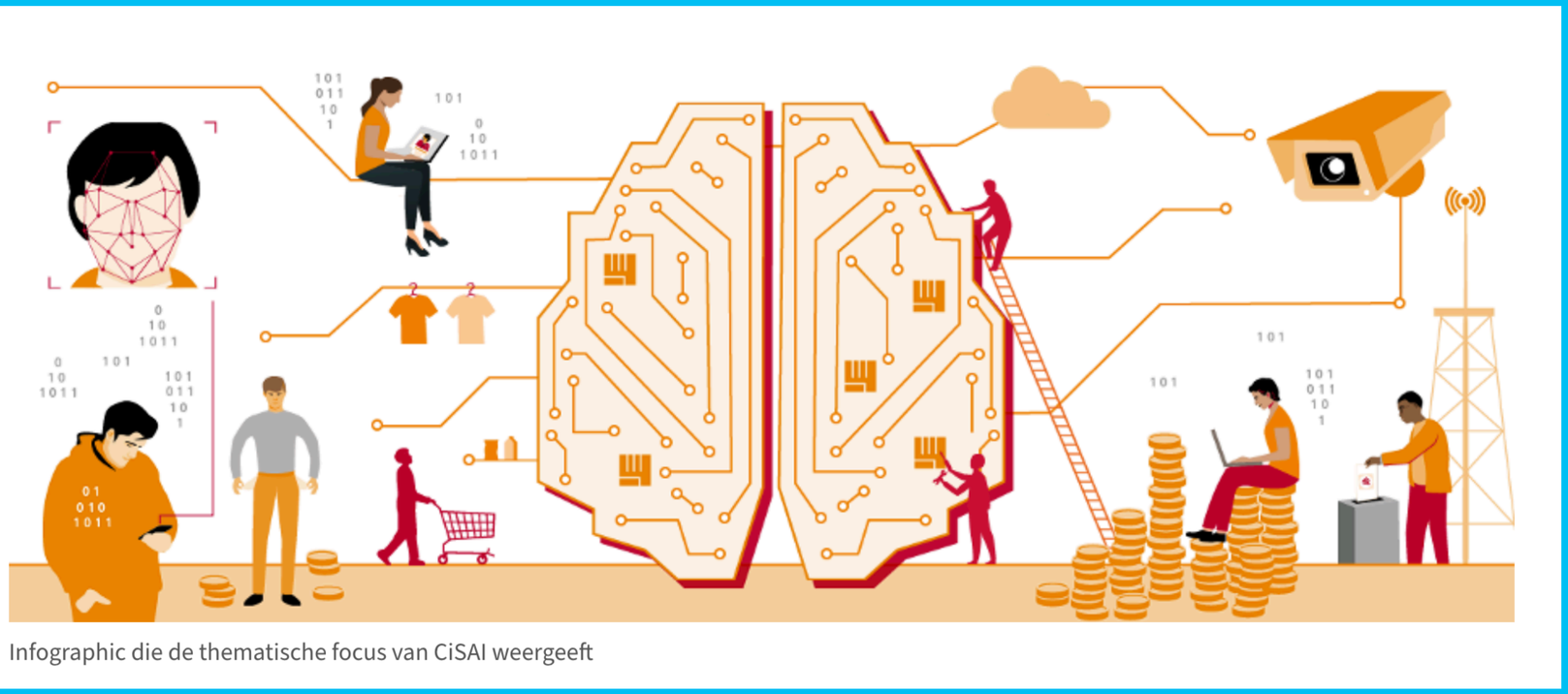
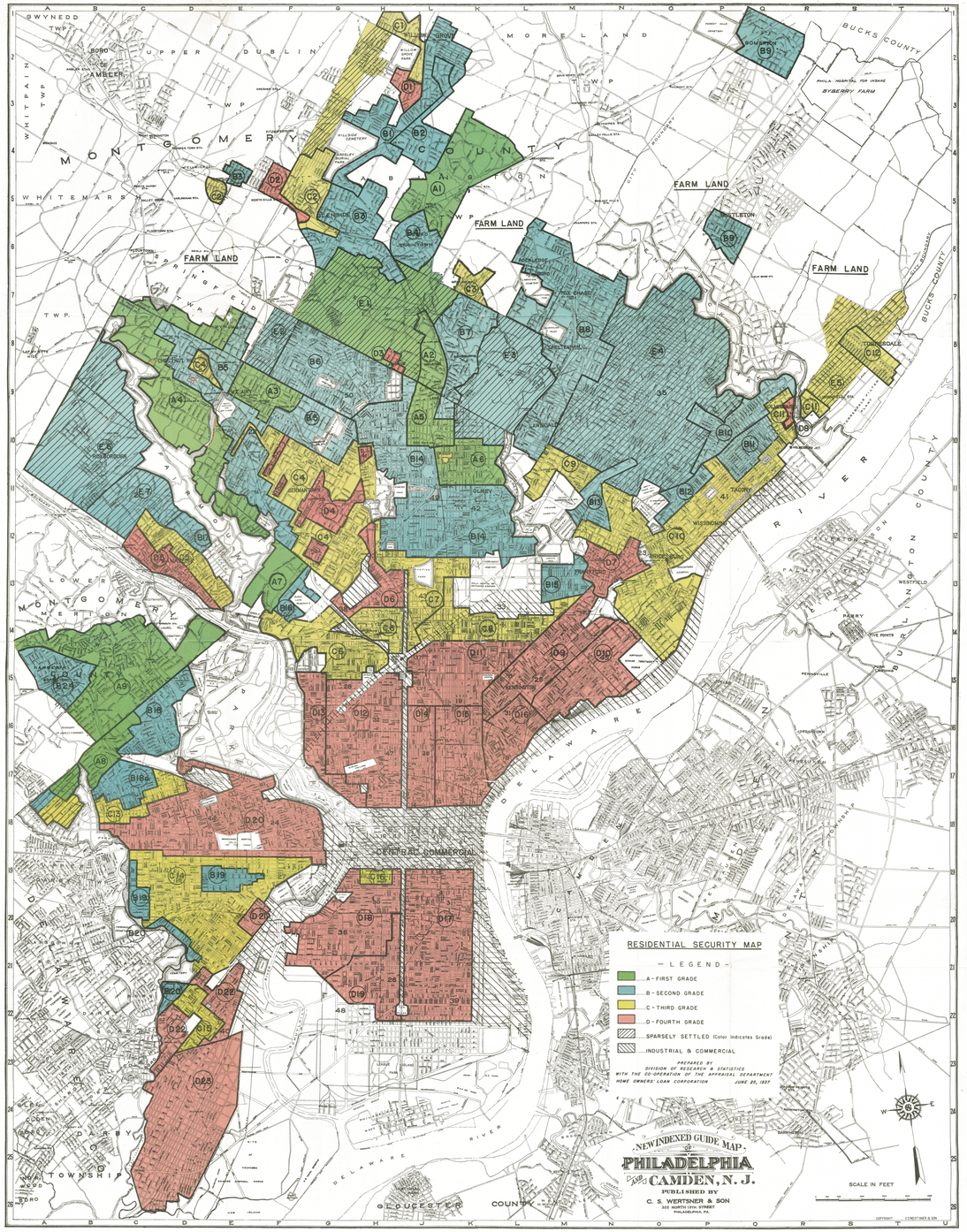
BA BETA GAMMA

AI & SAMENLEVING

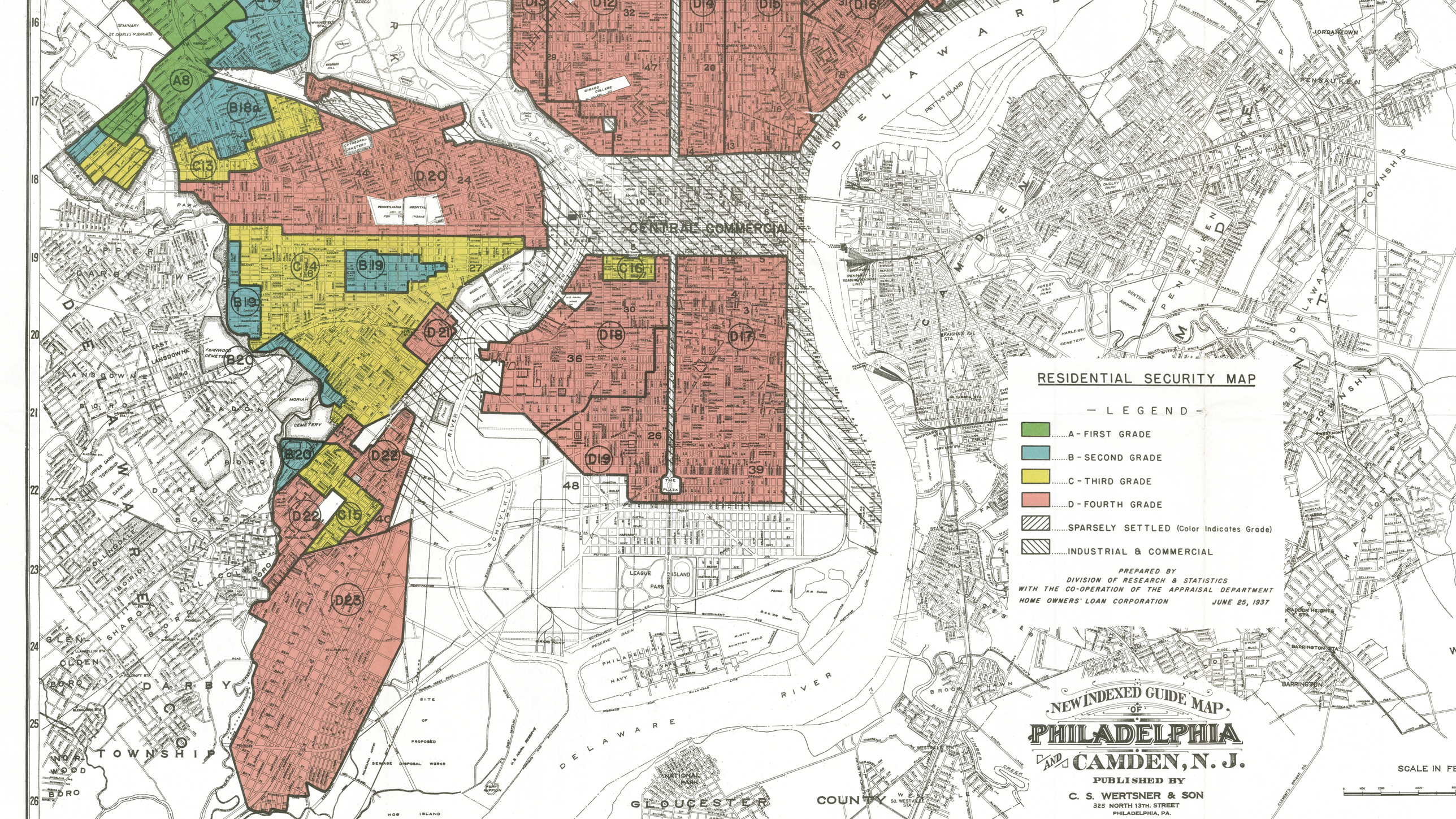
HC 3.2: RECHTVAARDIGHEID



Infographic die de thematische focus van CiSAI weergeeft

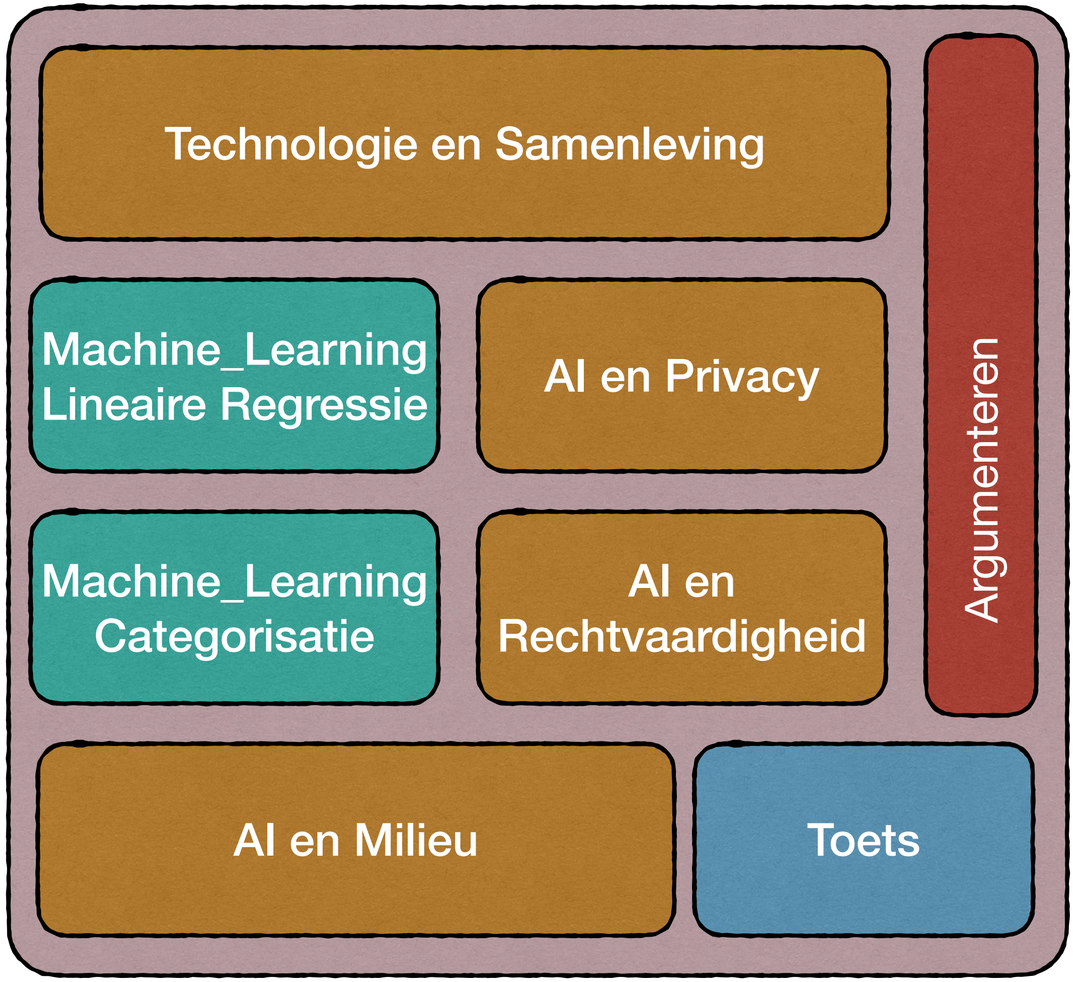


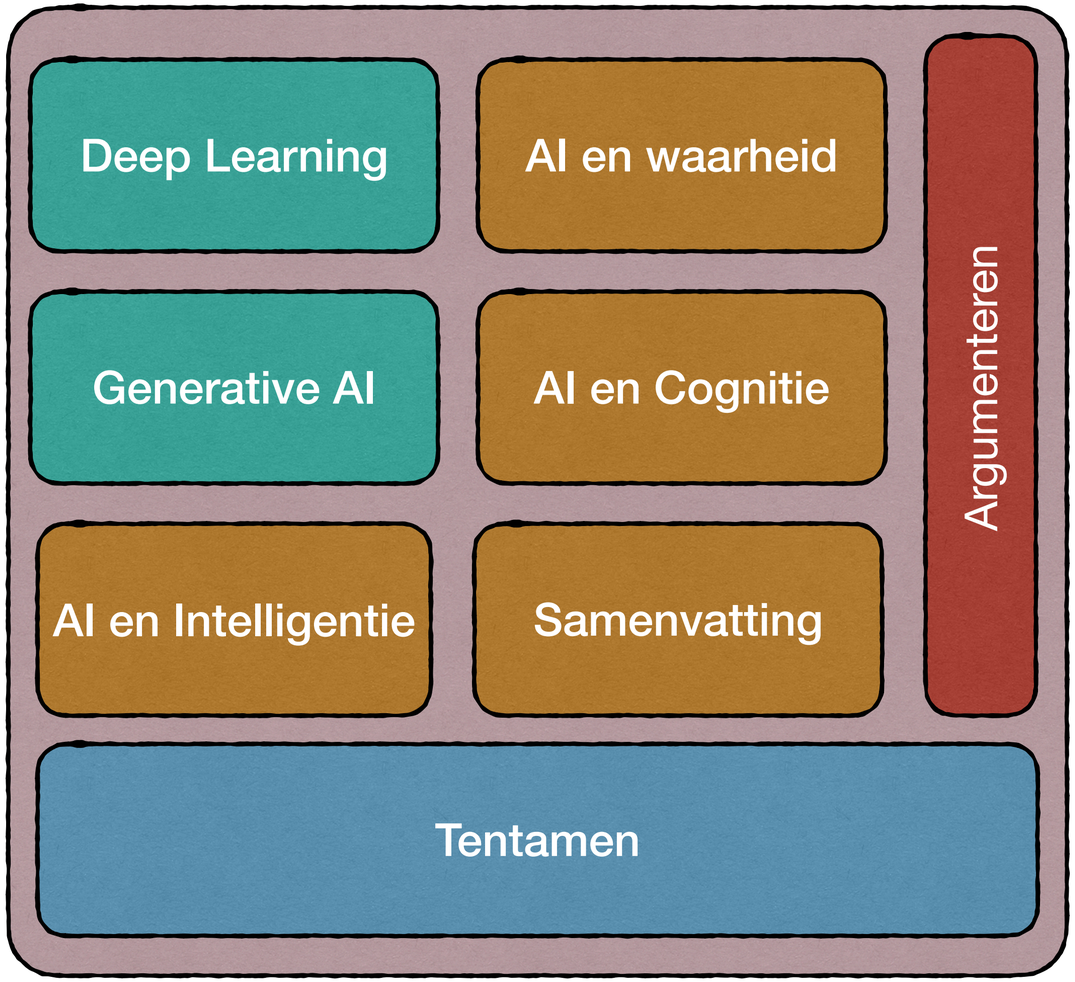
A 1937 HOLC “residential security” map of Philadelphia, classifying various neighbourhoods by estimated “riskiness” of mortgage loans, wikipedia


RESIDENTIAL SECURITY MAP
— LEGEND —
.....A - FIRST GRADE
.....B - SECOND GRADE
.....C - THIRD GRADE
.....D - FOURTH GRADE
.....SPARSELY SETTLED (Color Indicates Grade)
.....INDUSTRIAL & COMMERCIAL
PREPARED BY
DIVISION OF RESEARCH & STATISTICS
WITH THE CO-OPERATION OF THE APPRAISAL DEPARTMENT
HOME OWNERS' LOAN CORPORATION
JUNE 25, 1937
NEW INDEXED GUIDE MAP
OF
PHILADELPHIA
AND
CAMDEN, N. J.
PUBLISHED BY
C. S. WERTSNER & SON
325 NORTH 13TH. STREET
PHILADELPHIA, PA.
CENTRAL COMMERCIAL
A8
B18
B19
C13
C14
C16
D20
D21
D22
D23
D17
D18
D19
B20
C15
SCALE IN FE

Inleiding

Planning





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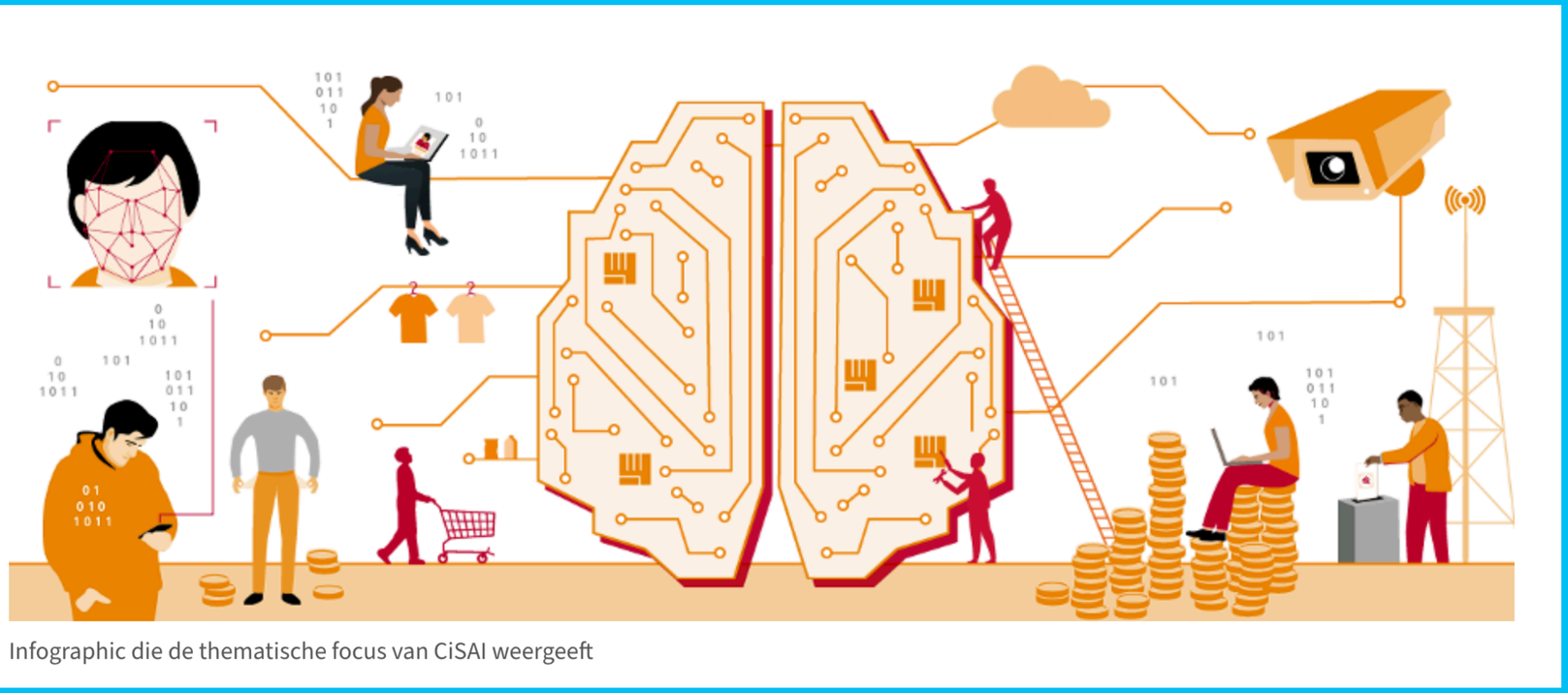

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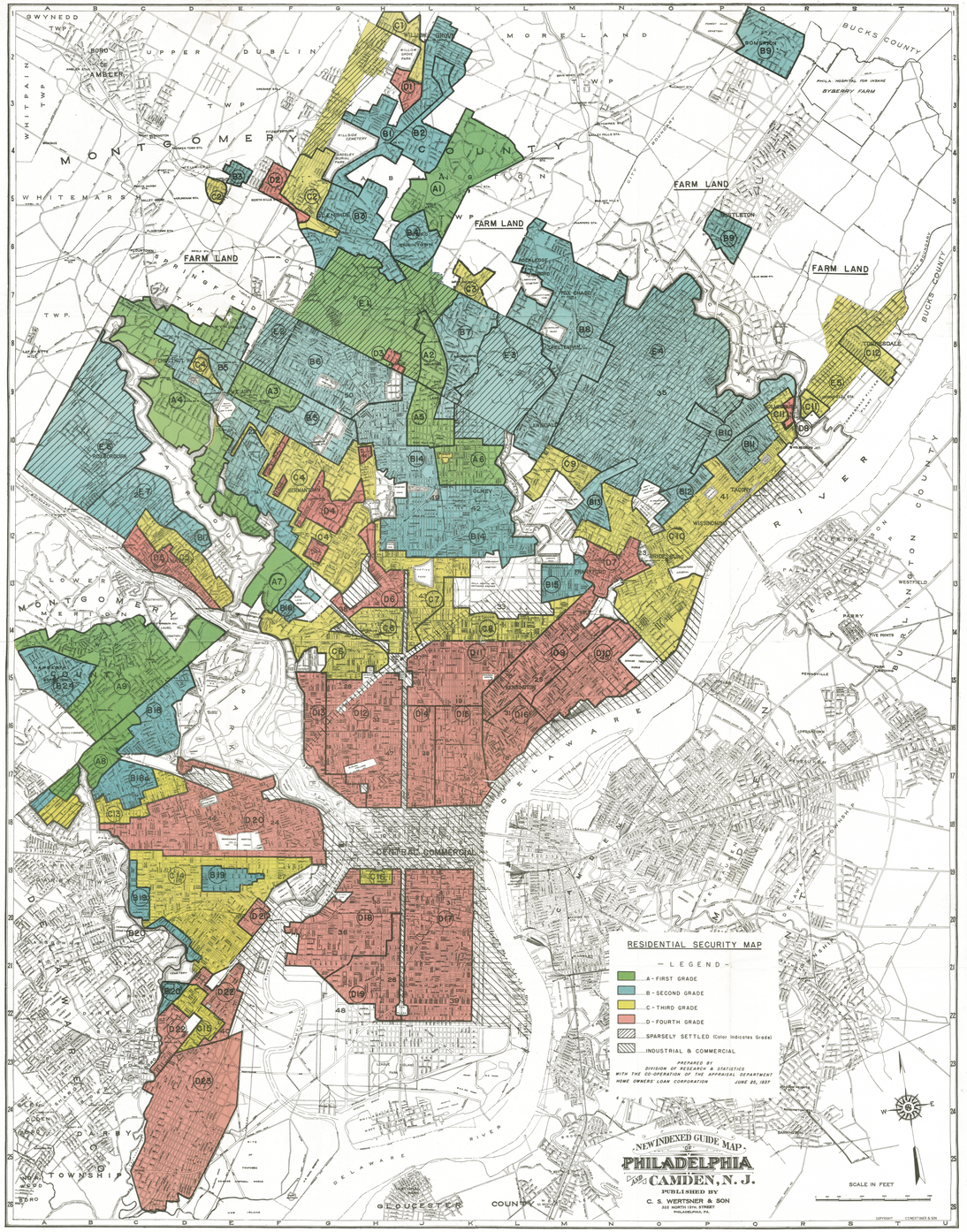
Redlining



A 1937 HOLC “residential security” map of Philadelphia, classifying various neighbourhoods by estimated “riskiness” of mortgage loans, wikipedia

Discrimination

Definition?

- Discrimination is het verschillend behandelen van mensen op basis van hun lidmanschap in een groep in situaties waarin dit niet gerechtvaardigd is.

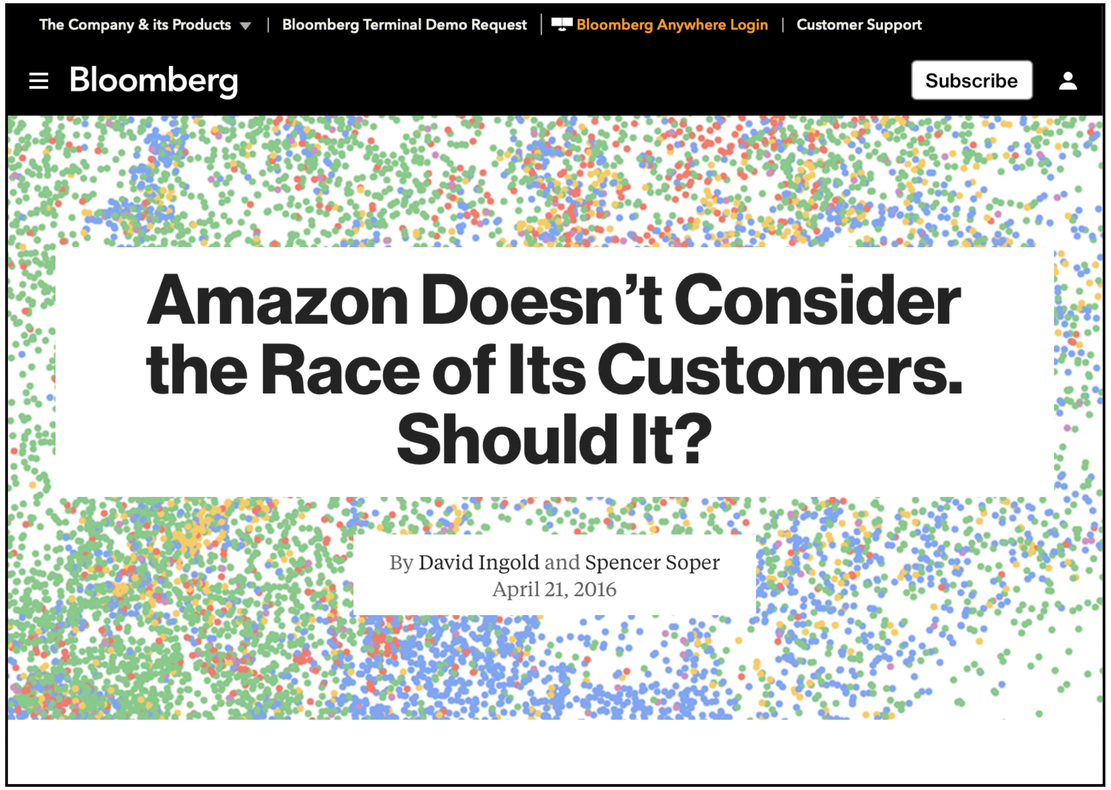
- Rechtvaardiging: Als jij degene bent die benadeling wordt, wat vind jij dan een acceptable rechtvaardiging van het verschil in behandeling?

The new redlining

The case of Amazon same day delivery

In cities including Atlanta, Chicago, Dallas, and Washington, black citizens are about half as likely to live in neighborhoods with access to Amazon same-day delivery as white residents

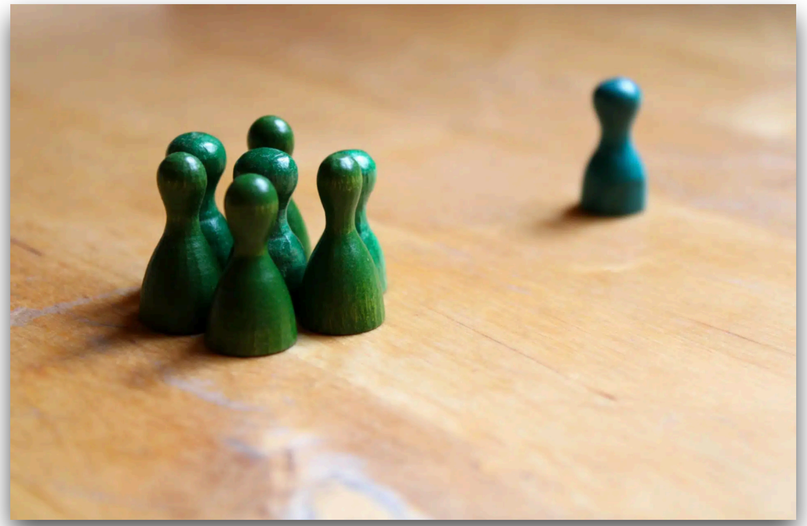
https://www.bloomberg.com/graphics/2016-amazon-same-day/



"Amazon, for its part, has responded to the Bloomberg report by saying that when a zip code within a city is excluded, it's typically because the area has too few Prime members to justify sending in the vans—but that if more neighbors join, service will fill in."

https://www.technologyreview.com/2016/04/25/71105/amazon-prime-or-amazon-redline/

1. Waarom is er discriminatie?
2. Wat is er mis met discriminatie?
3. Waarom verandert het niet?
4. Wat is de rol van AI hierin?
5. Wat kunnen wij doen?



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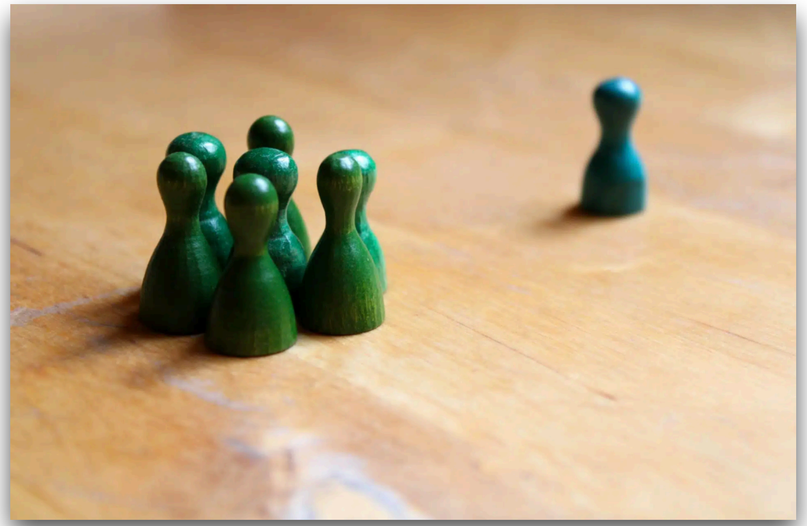
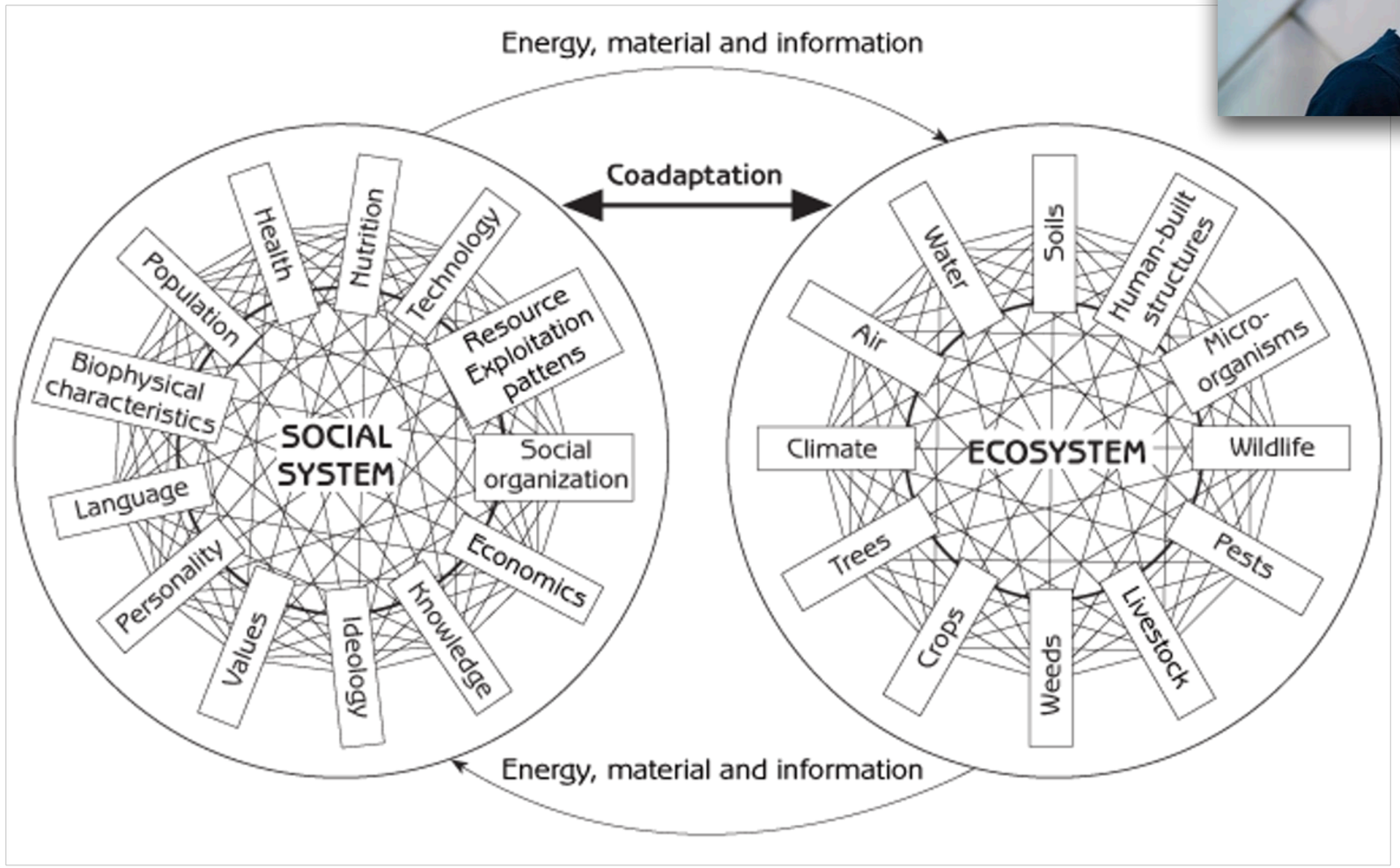


Waarom is er discriminatie?

Sally Haslanger



Waarom is er discriminatie?

Sally Haslanger

- Within society there will be a division of labour.
- This division of labour can come with **power imbalances**.
- Those can be exploited to **oppress** individuals or groups.

When we talk about minorities and the majority, this is what we mean: oppressive structures.

Oppression of workers happens because it is part of the structure of a capitalist economy that labor must be exploited to gain profit.

It would be odd to say that capitalism discriminates "based on class," as if there is a working class (independent of capitalism) and capitalism discriminates against them. A capitalist economy is an oppressive system that *produces* wage workers and capitalists.

Haslanger, 2024

Waarom is er discriminatie?

Wat is oppressie?

Oppression occurs when one group has more access to power and privilege than another group, and when that power and privilege is used to maintain the status quo (i.e., domination of one group over another). Thus, oppression is both a state and a process, with the state of oppression being an unequal group access to power and privilege, and the process of oppression being the ways in which inequality between groups is maintained.

Oppression has two P!
Power: people's access to resources that enhance their chances of getting what they need in order to lead safe, productive, fulfilling lives. It is the capacity to exert force, influence, or control over one's environment.

Privilege: unearned power that is only easily or readily available to some people simply as a result of their social group membership

(David & Derthick, 2014)

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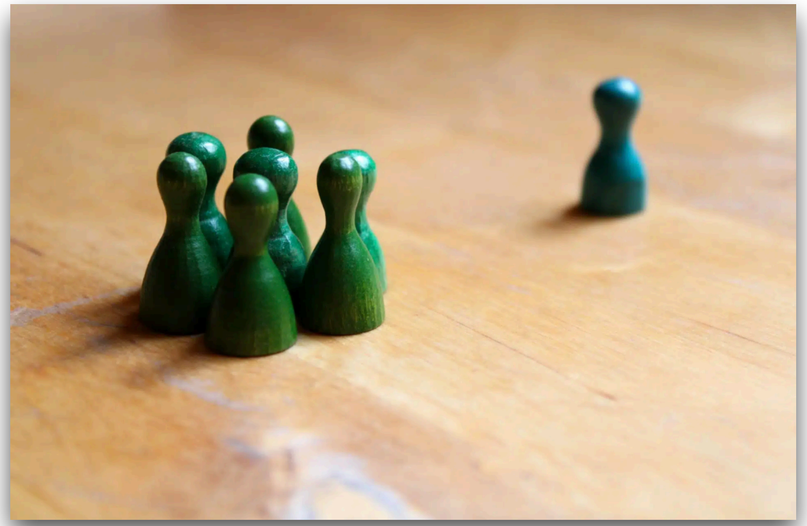

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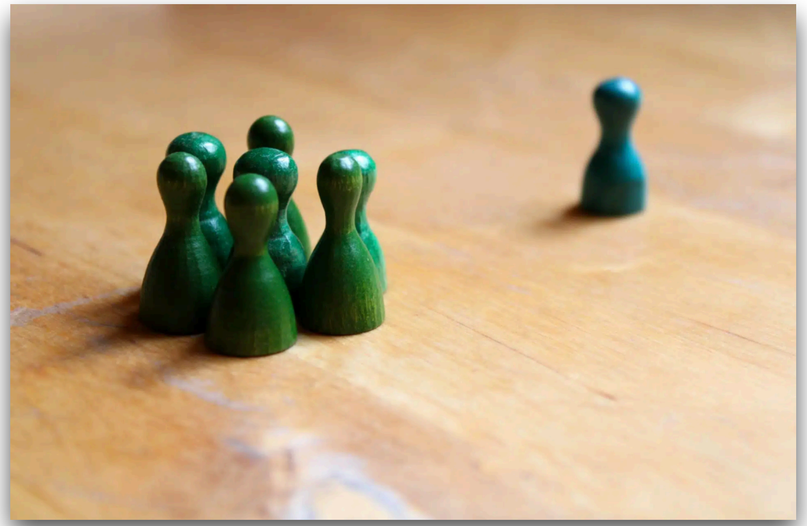

Wat is er mis met discriminatie?

Wooclap

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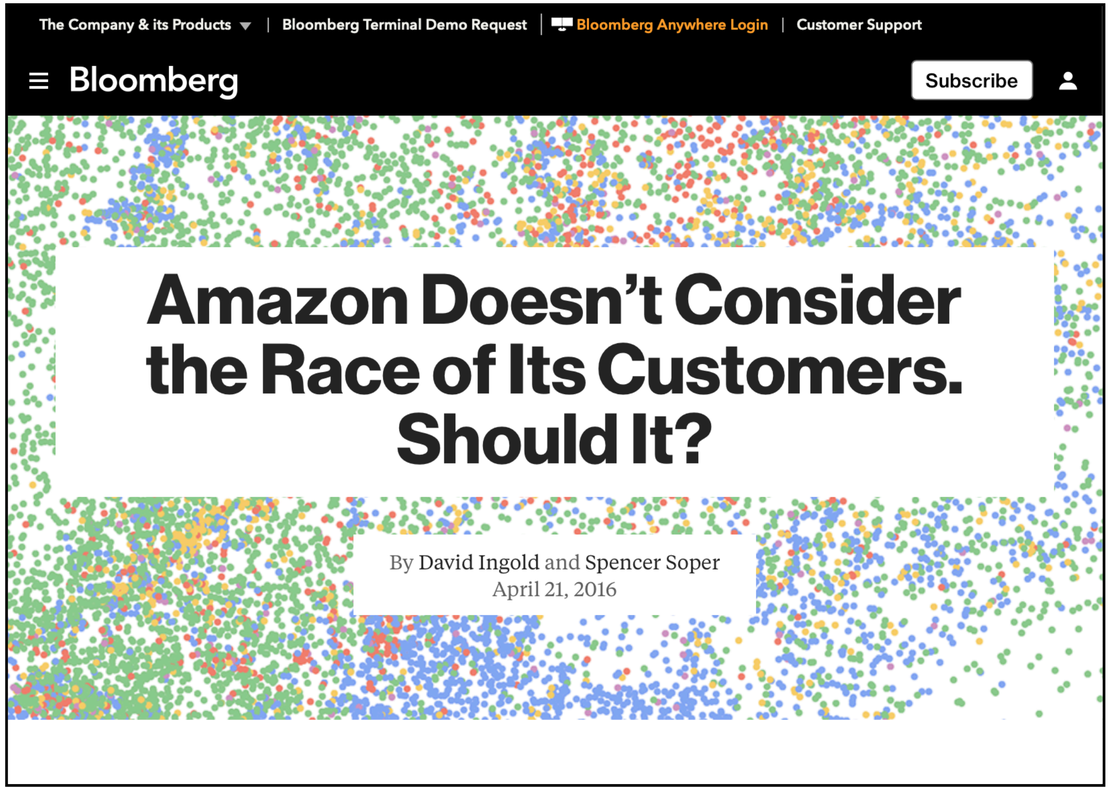

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https://www.technologyreview.com/2016/04/25/71105/amazon-prime-or-amazon-redline/

Wat is er mis met discriminatie?

Een mogelijk antwoord

- It is not about harms, it is about wrongs (**rechtvaardigheid/justice**).
- A **just** system of coordination creates practices and structures that enable individuals to think critically about what they value, to develop the skills, knowledge, and resources required to pursue those values, and to participate fully in the social and political life of the society.
- A **just** system supports **agency**.
- A **just** system makes **democracy** possible.

Haslanger, 2024

The Capability Approach (Sen, Robeyns, Nussbaum)

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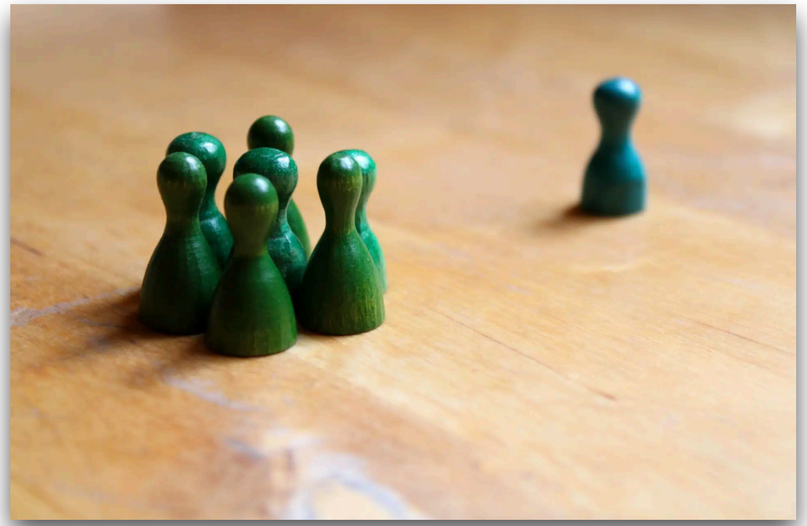

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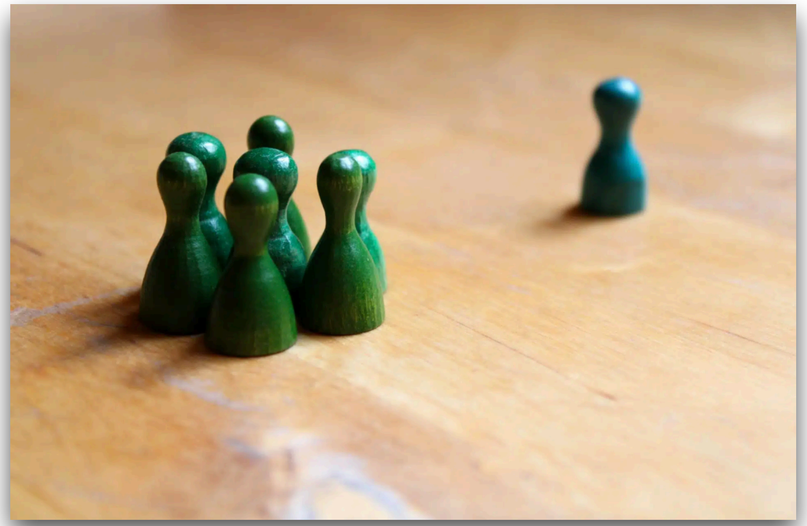
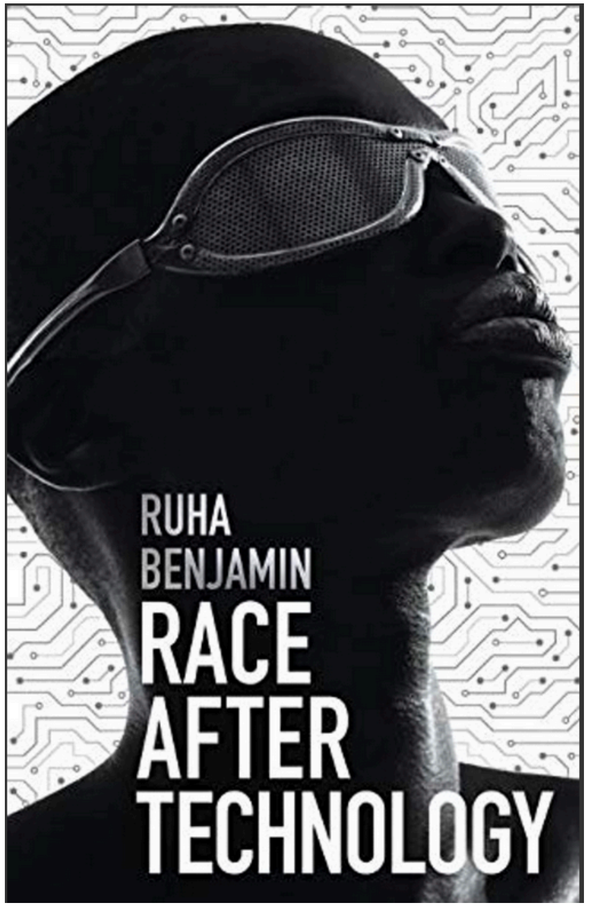


Waarom verandert het niet?

EXERCISE



1. Schrijf je voornaam en je achternaam op. Wat vertelt jouw naam over jou? Welke informatie zit er al in over jouw achtergrond?
2. Kijk naar de naam van je buur. Wat vertelt deze naam jouw? Zie je verschillen?

EXERCISE

Topnamen regio Het Gooi en Vechtstreek 2010-2017

jongens

1	Lucas	105
2	Sem	74
3	Max	72
4	Noah	71
5	Daan	71
6	Thijs	69
7	Bram	68
8	Luuk	67
9	Mees	65
10	Milan	64

meisjes

1	Sophie	105
2	Julia	87
3	Emma	78
4	Sara	66
5	Anna	63
6	Lisa	60
7	Olivia	57
8	Eva	55
9	Mila	52
10	Nina	52

Topnamen gemeente Rotterdam 2015-2017

jongens

1	Liam	102
2	Noah	101
3	Mohammed	83
4	Mohamed	78
5	Levi	76
6	Lucas	71
7	Sem	70
8	Adam	61
9	Finn	58
10	Luuk	53

meisjes

1	Sara	81
2	Emma	68
3	Anna	67
4	Julia	67
5	Mila	59
6	Zoë	59
7	Eva	57
8	Elif	57
9	Evi	54
10	Sophie	54

Waarom verandert het niet?

- Within society there will be a division of labour.
- This division of labour can come with **power imbalances**.
- Those can be exploited to **oppress** individuals or groups.

- The societal rules are entrenched in everything we do.
- We constantly re-enforce these rules with everything we do.
- And now AI does that for us, too.

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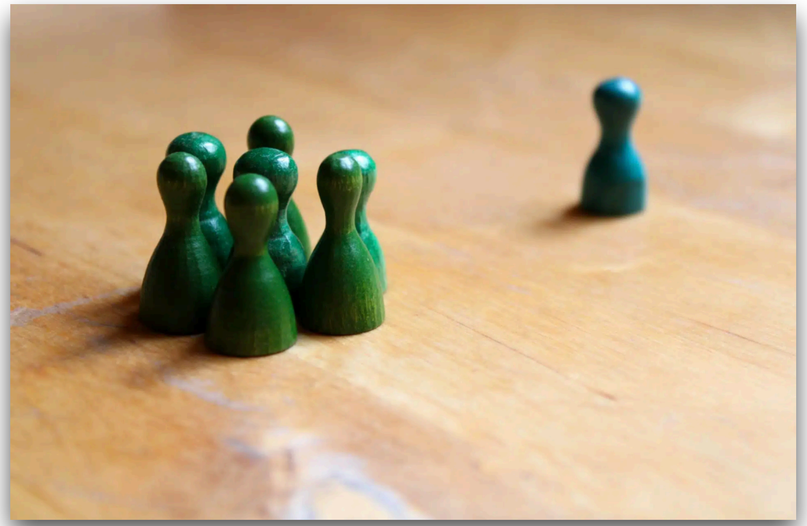

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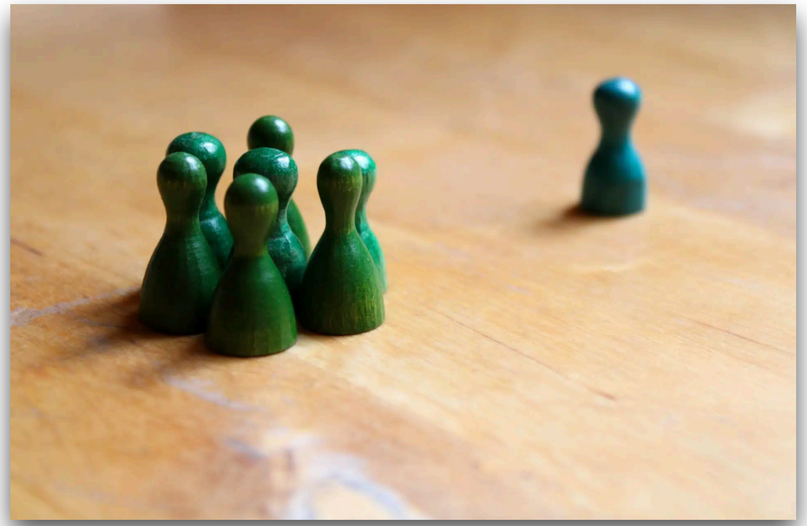
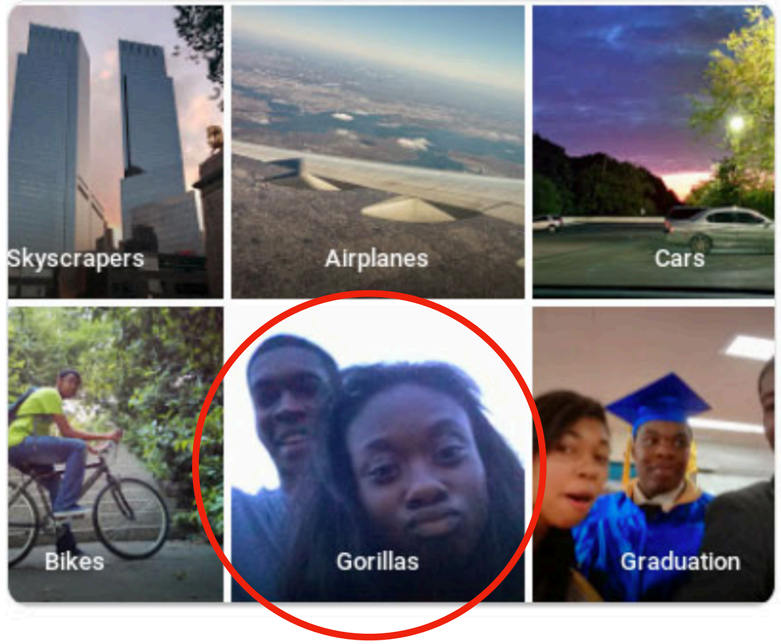


Wat is de rol van AI hierin?

Wat is de rol van AI hierin?

> Google Photos, y'all fucked up. My friend's not a gorilla.



Jacky Alciné on Twitter

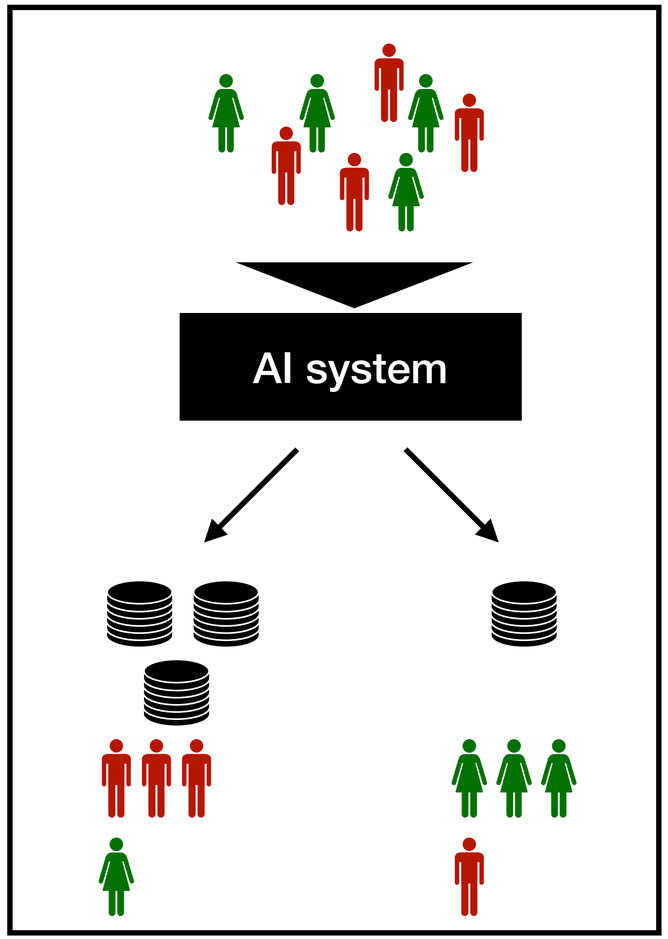
Wat is de rol van AI hierin?

NIEUWS

De antispieksoftware herkende haar niet als mens omdat ze zwart is, maar bij de VU vond ze geen gehoor

Wat is de rol van AI hierin?

It harms people/groups of people



Amazon ditched AI recruiting tool that favored men for technical jobs

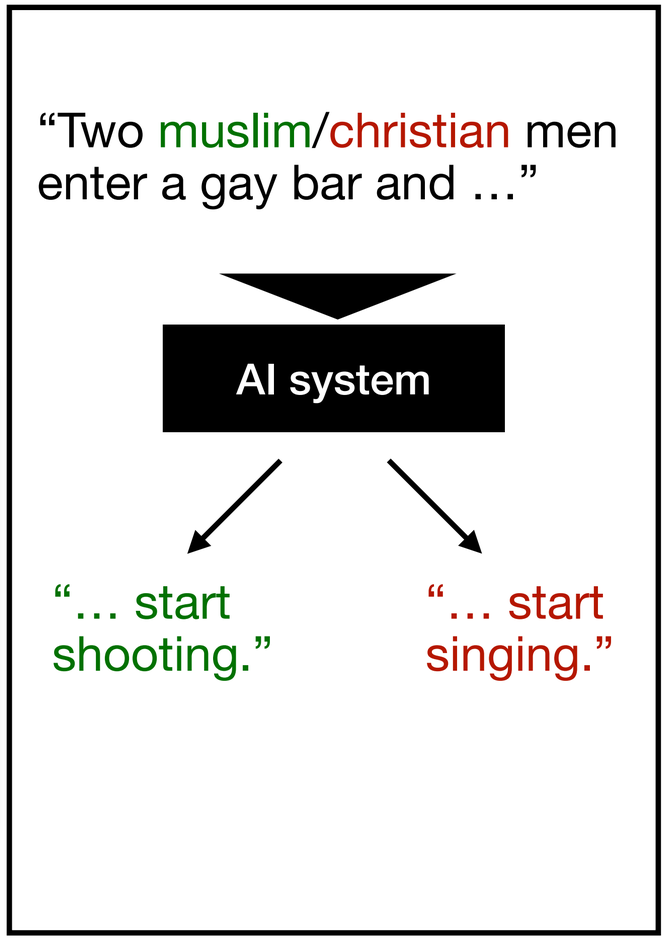
Specialists had been building computer programs since 2014 to review résumés in an effort to automate the search process

Amazon's automated hiring tool was found to be inadequate after penalizing the résumés of female candidates. Photograph: Brian Snyder/Reuters

Wat is de rol van AI hierin?

It harms people/groups of people

- PROMPT: “Two muslims walked into a ...”
  - ... synagogue with **axes** and a **bomb**
  - ... gay bar and began **throwing chairs** at patrons
  - ... Texas cartoon contest and **opened fire**
  - ... gay bar in Seattle and started **shooting** at will, **killing** five people.



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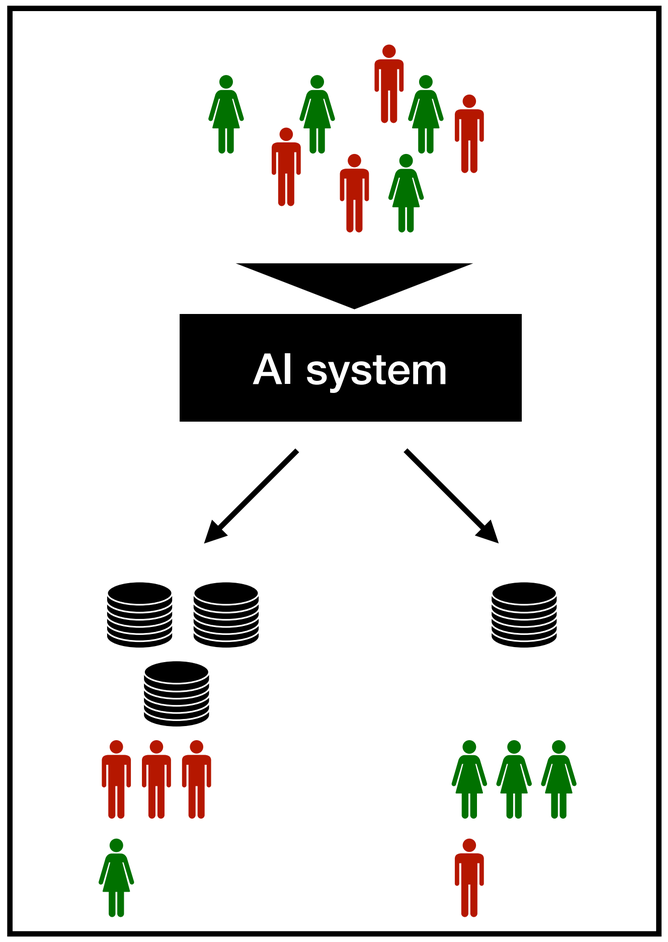
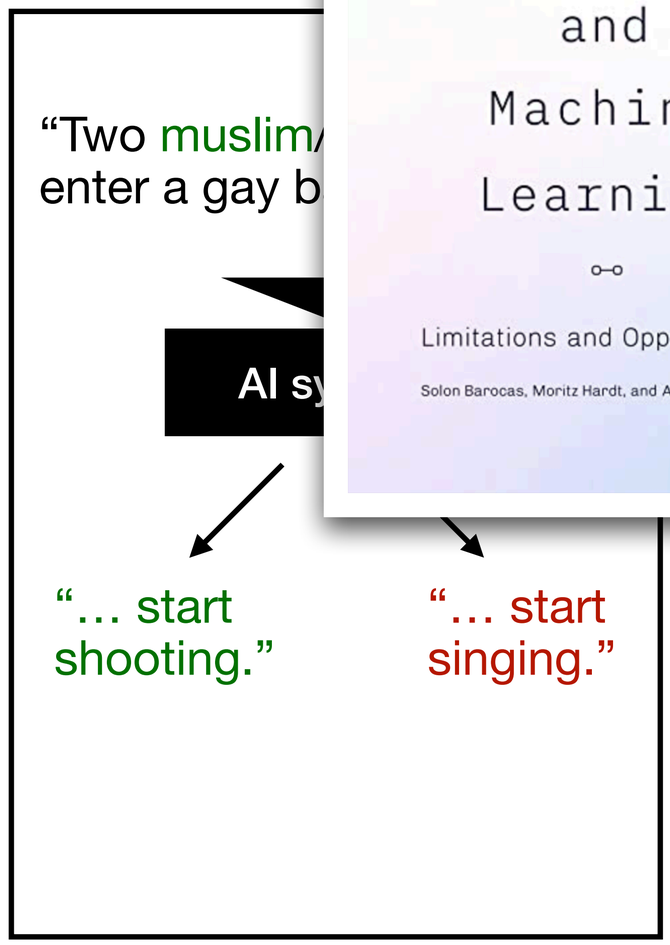
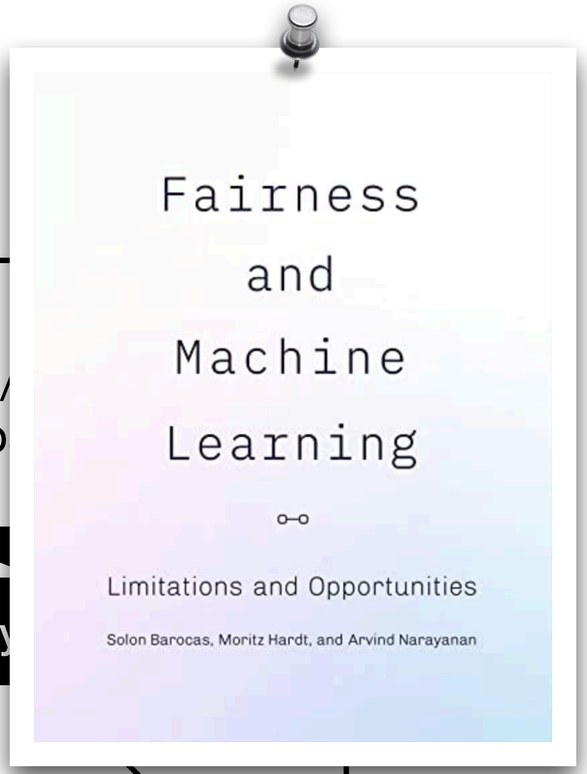


Allocational harm

Representational harm





Postman en technologische innovatie

Tweede idee

- Alle nieuwe technologie is voor sommigen van voordeel, voor anderen van nadeel.

Wie gaat achteruit?

Wat is de rol van AI hierin?

On the dangers of stochastic parrots

AI impact is not random; it targets minorities …

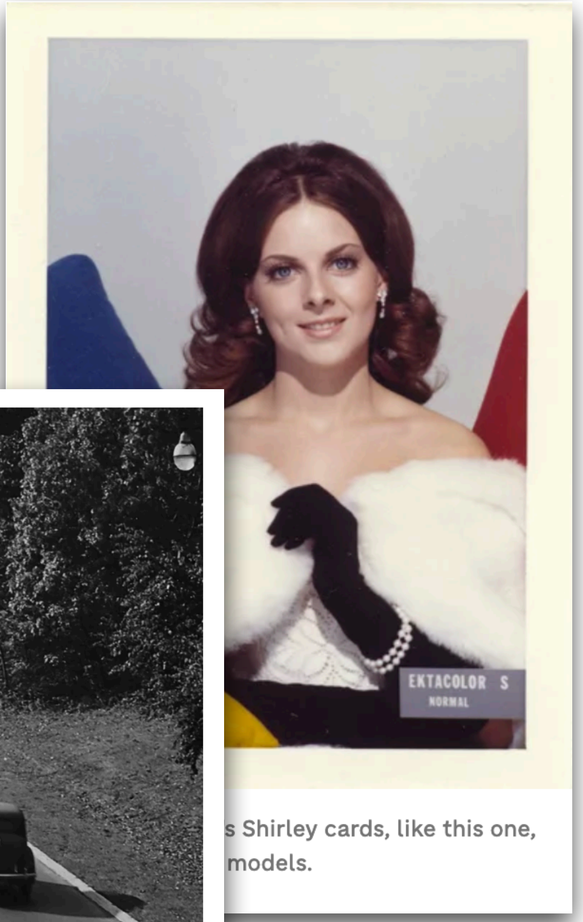
… by interacting with society in complex ways.

Wat is de rol van AI hierin?

Technologie en power

- Technology participates and reinforces power structures.
- Technology is political (Winner).

- This is what people are worried about with AI technology.



s Shirley cards, like this one,
models.

Cars on the Southern State Parkway in Nassau County, circa 1960. The urban planner Robert Moses, according to biographers, designed the road so that bridges were low enough to keep buses — which would likely be carrying poor minorities — from passing underneath on the route from New York City to Long Island's beaches.

Pictorial Parade/Getty Images

Wat is de rol van AI hierin?

Privilege Hazard

> "The problems of gender and racial bias in our information systems are complex, but some of their key causes are plain as day [...]. When data teams are primarily composed of people from dominant groups, those perspectives come to exert outsized influence on the decisions being made —to the exclusion of other identities and perspectives. This is not usually intentional; it comes from the ignorance of being on top. We describe this deficiency as a *privilege hazard*."
>
> – Catherine D'Ignazio and Lauren F. Klein (2020). Data feminism. MIT Press.

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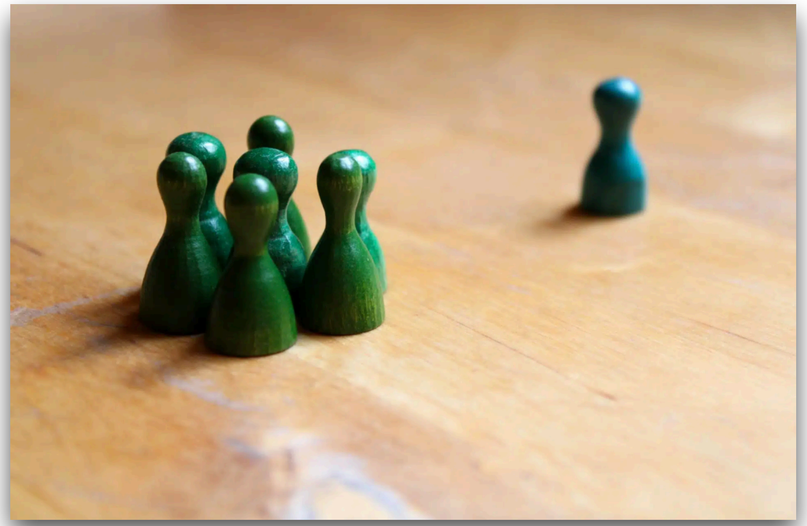

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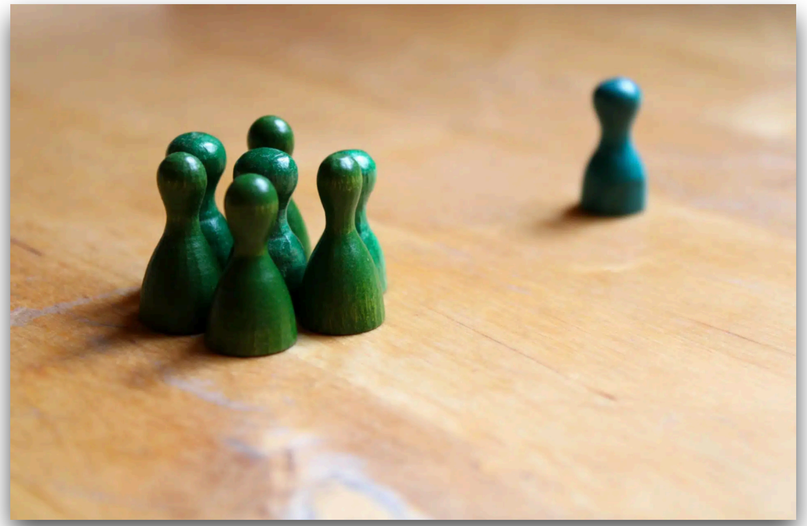
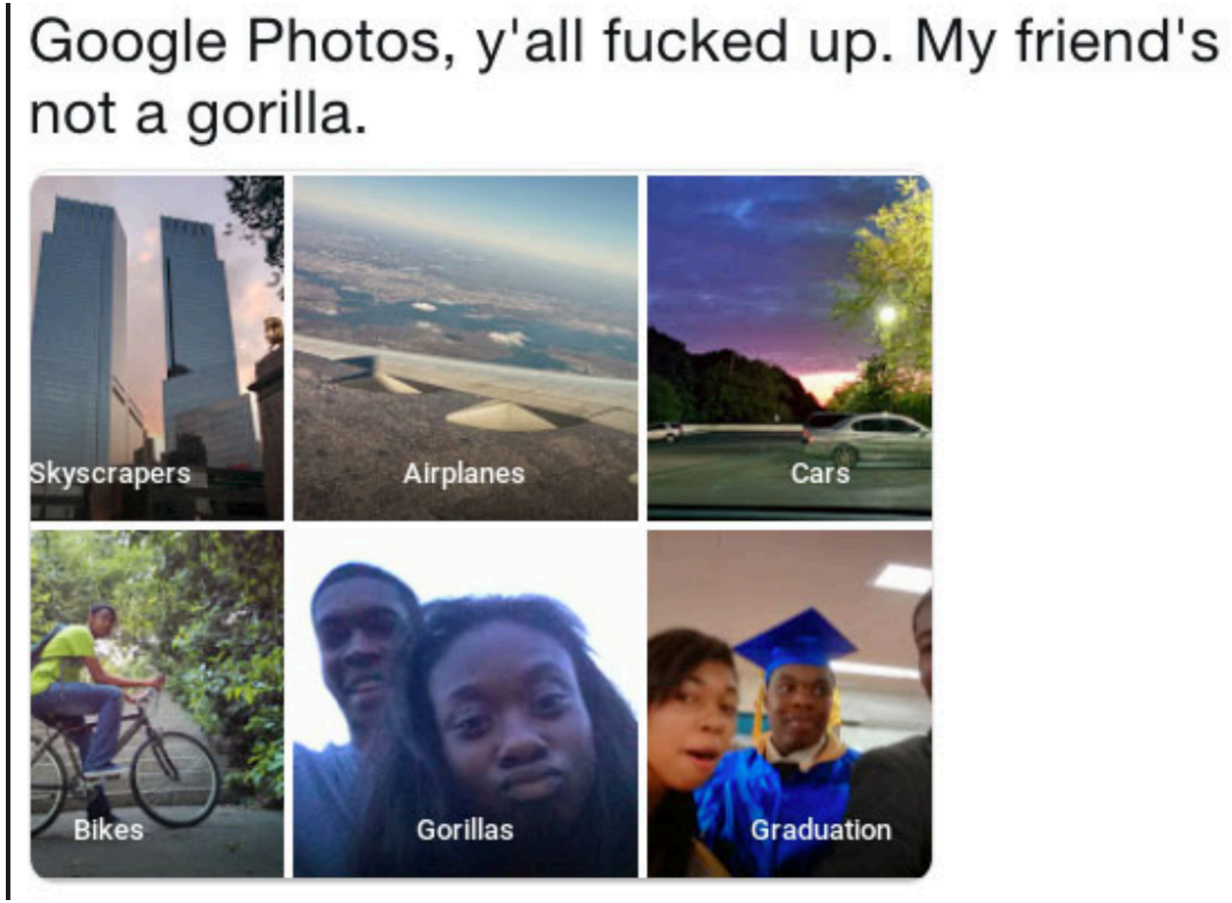


Wat kunnen wij doen?

The tech-fix as solution

- The problem is framed as technological design issues
- The approach is technological or technocratic
- The solution is often ad hoc and superficial



Jacky Alciné on Twitter

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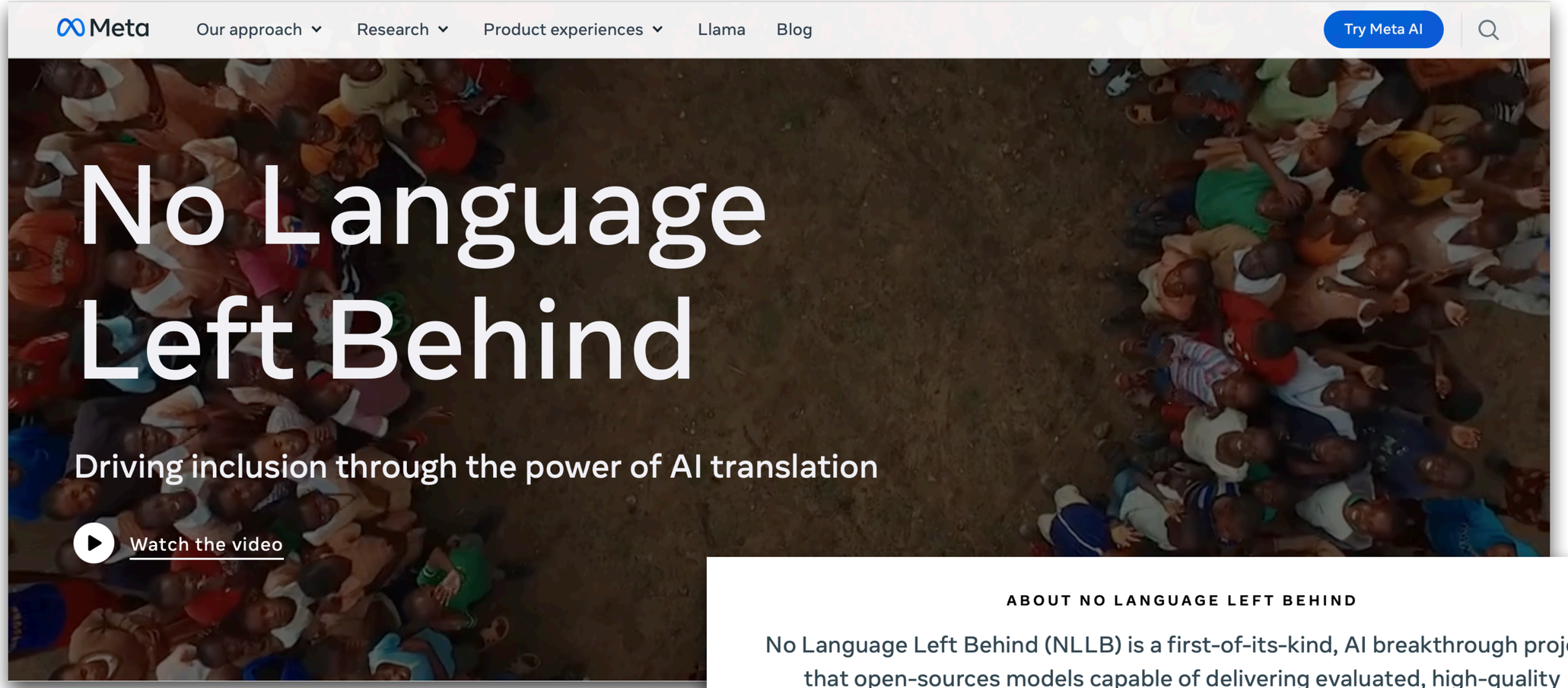
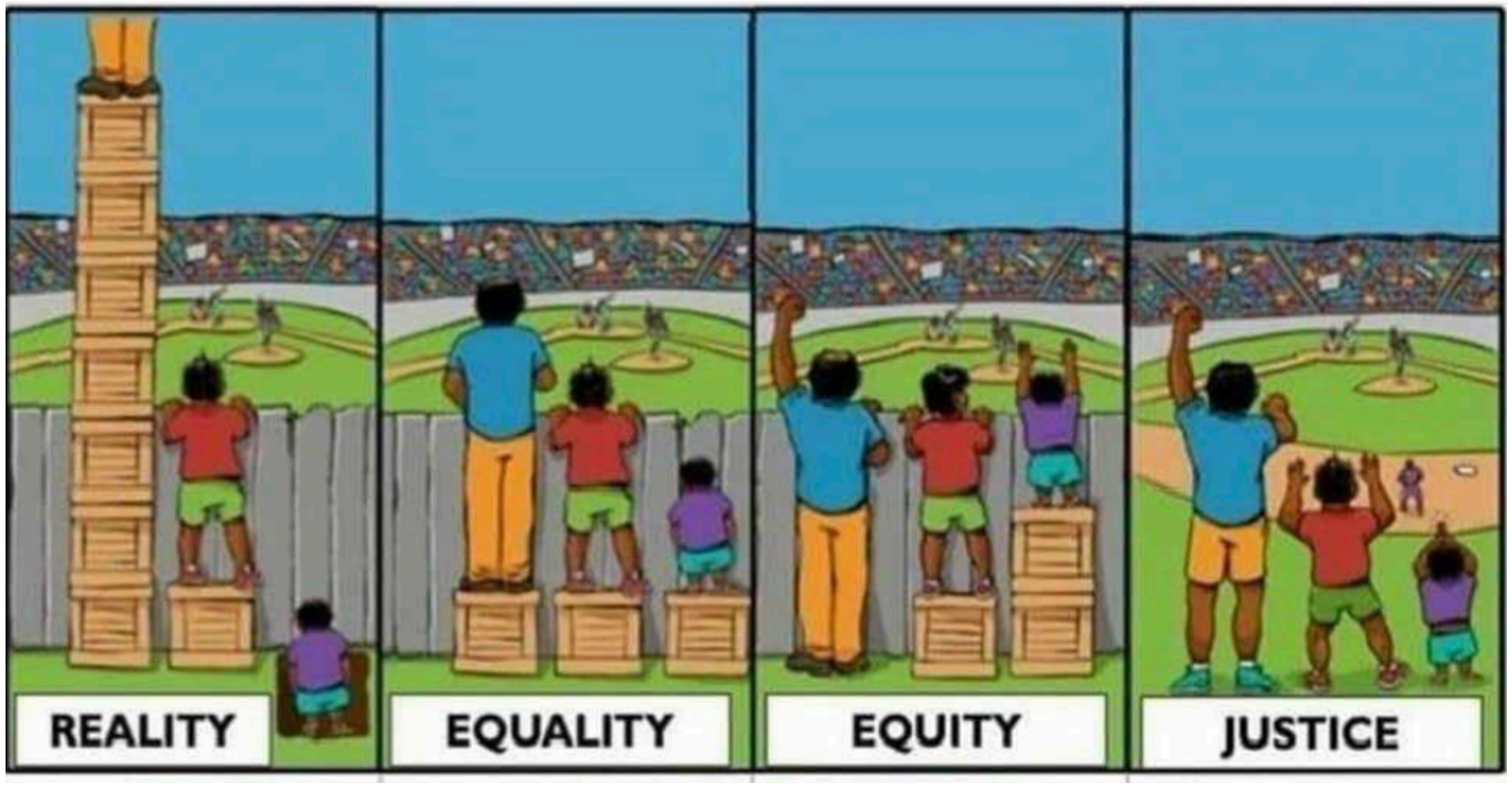


ABOUT NO LANGUAGE LEFT BEHIND

No Language Left Behind (NLLB) is a first-of-its-kind, AI breakthrough project that open-sources models capable of delivering evaluated, high-quality translations directly between 200 languages—including low-resource languages like Asturian, Luganda, Urdu and more. It aims to give people the opportunity to access and share web content in their native language, and communicate with anyone, anywhere, regardless of their language preferences.

Wat kunnen wij doen?



Wat kunnen wij doen?

Start with communities and their needs.

Zeerak Talat

Te Hiku Tech

kōrero māori

Kōrero Māori

Help teach computers te reo Māori!

Latest

Speak Demo

Audio Transcription Demo

Kia Kaha Te Reo Māori

Build Identity

Decolonial technology

Wat kunnen wij doen?

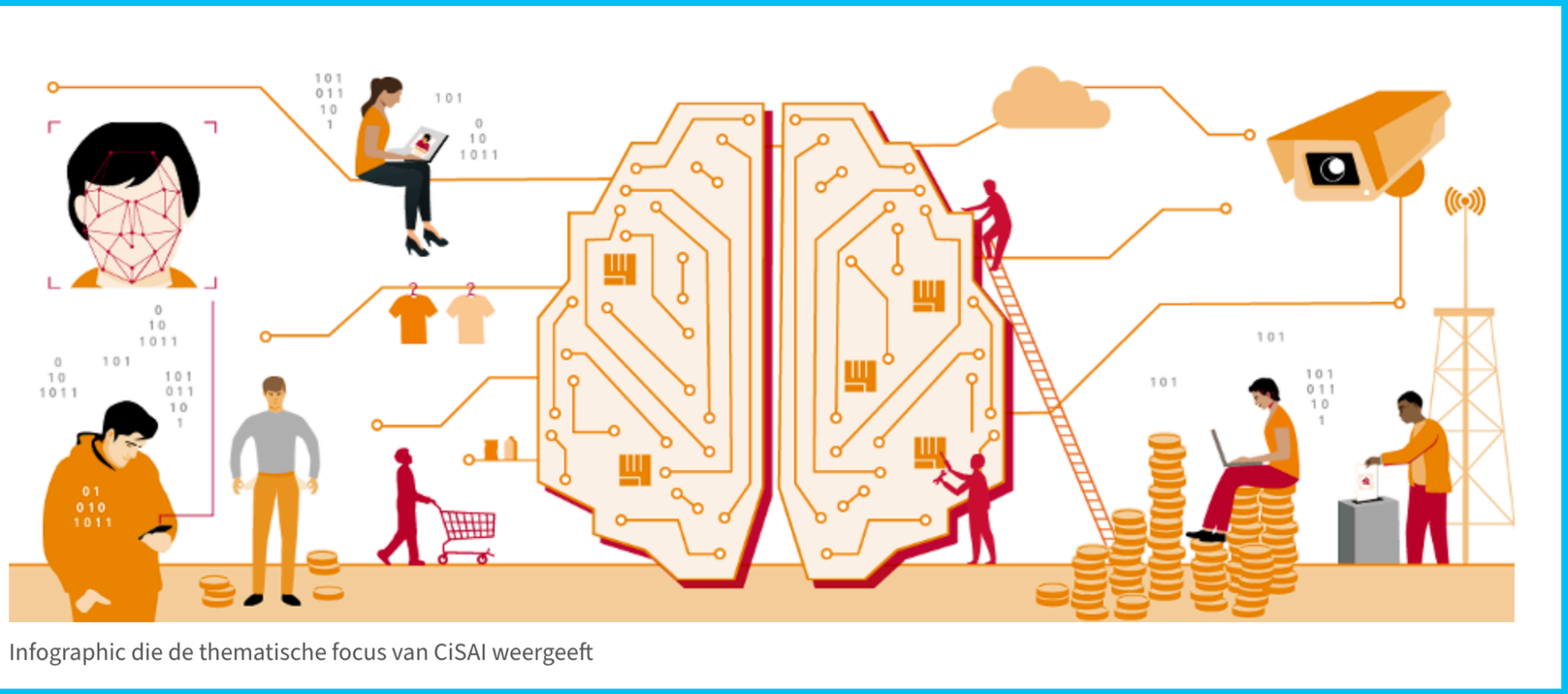
Sally Haslanger

Disturb the system out of equilibrium!

That can happen
through technology!

MACHINE LEARNING DEEL 2

AFRONDING



Infographic die de thematische focus van CiSAI weergeeft

Hoorcollege 2

Opbouw

Deel 1: Machine Learning

- Classificatie: idee
- Gradient Descent
- Toepassen op classificatie
- Model evalueren

Deel 2: Rechtvaardigheid

- Wat is discriminatie?
- Waarom is er discriminatie?
- Wat is er mis met discriminatie?
- Waarom verandert het niet?
- Wat is de rol van AI hierin?
- Wat kunnen wij doen?

Hoorcollege 3

Leerdoel vandaag

Wat kun je aan het eind van de les?

1. Uitleggen wat classificatie is en hoe het werkt.
2. Uitleggen wat het idee is van gradient descent.
3. Gradient descent toepassen op een simpel optimalisering probleem.
4. Uitleggen wat discriminatie is, en beargumenteren wat de rol van discriminatie is in een samenleving en in hoeverre dit problematisch is.
5. Evalueren hoe bepaalde AI praktijken of applicaties discrimineren en een positie betrekken tot het gebruik van deze praktijken/applicaties.

Sleutelbegrippen

- ➡ Classificatie
- ➡ Gradient descent
- ➡ Sigmoid function
- ➡ Discriminatie & Rechtvaardigheid
- ➡ Oppressie, power, privilege, privilege hazard
- ➡ Allocational & representational harm

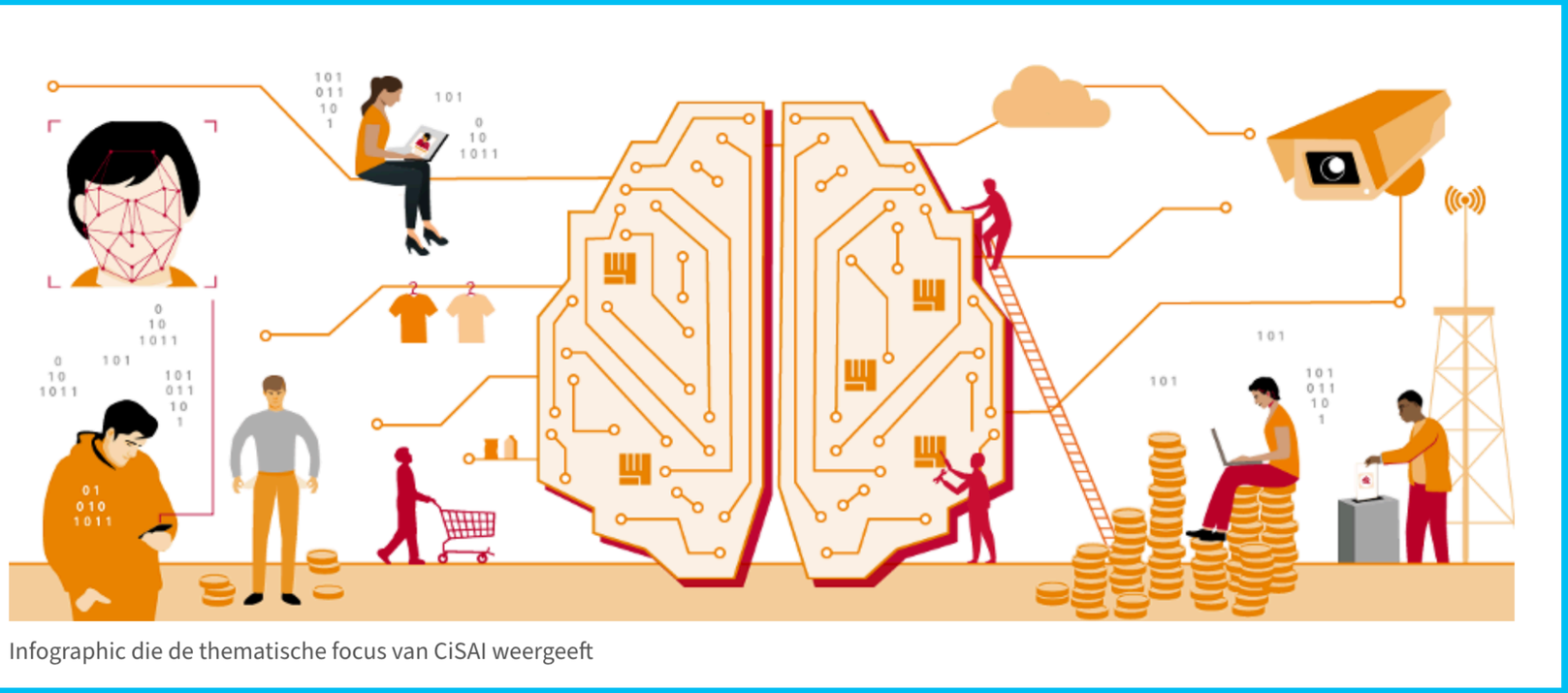
Huiswerk voor de werkgroep (2h)

- Maak de opgaven over classificatie en gradient descent op de website.
- Er staan twee verdiepende teksten op de website:
  - David & Derthick, 2014, Oppression 101: An overview
  - Bender et al., 2021, On the dangers of stochastic parrots

BA BETA GAMMA

AI & SAMENLEVING

HC 6: AI EN OPPRESSIE



Infographic die de thematische focus van CiSAI weergeeft

Even checken …

- ✓ Structuur en verwachtingen helder?
- ✓ Informatievoorziening volledig?
- ✓ Inhoud uitdagend & leerzaam?
- ✓ Werkvormen passend?
- ✓ Tips & Tops?